Ce Zhang

TEST SAMPLE (fictional) | Mechanical Design Engineer

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EDUCATION

East China Test University of Technology (fictional) Sep 2025 – Mar 2028 (expected)

Master of Engineering in Mechanical Engineering

- Research focus: gear transmission dynamics and lightweight design

East China Test University of Technology (fictional) Sep 2021 – Jun 2025

Bachelor of Engineering in Mechanical Design, Manufacturing and Automation

- GPA 3.52/4.00; core courses: Theory of Machines, Machine Design, Engineering Mechanics, Hydraulic and Pneumatic Transmission

PROJECT EXPERIENCE

Modal Analysis and Lightweight Design of a Marine Gear Reducer Housing Oct 2025 – Present

Research Team Member

- Built the housing model in SolidWorks and ran a free modal analysis of the first 6 modes in ANSYS Workbench; identified 2 resonance-risk regions
- Ran topology optimization on the rib layout; reduced housing mass from 48.6 kg to 43.1 kg while keeping the first natural frequency at or above the baseline
- Specified key mounting-face flatness ≤ 0.02 mm, bearing bore Φ90H7 and surface roughness Ra 1.6 μm per GB/T 1184 on the drawings

PLC-Controlled Material Sorting Conveyor Mar 2024 – Jun 2024

Team Lead (3 students), Course Project

- Programmed ladder logic on a Siemens S7-1200 in TIA Portal to sort parts by color and size sensor signals
- Designed the pneumatic pusher and its circuit diagram; ran 30 minutes without jamming at the course acceptance test

PROFESSIONAL EXPERIENCE

Test Heavy Equipment Co., Ltd. (fictional) Jul 2024 – Sep 2024

Mechanical Design Intern

- Modeled conveyor support frames in SolidWorks and dimensioned and toleranced 12 part drawings
- Checked welded-part dimensions on the shop floor with a process engineer and compiled a list of common assembly issues

Mechanical Innovation Society (student club) Sep 2022 – Jun 2023

Head of Technical Team

- Ran 4 introductory SolidWorks workshops and led 6 members in assembling a competition prototype

TECHNICAL SKILLS

Design Software: SolidWorks, AutoCAD, ANSYS Workbench

Programming & Simulation: Python, MATLAB/Simulink, Siemens S7-1200 PLC

Engineering Fundamentals: GD&T

Languages: English: CET-6 (score 532)

CERTIFICATIONS AND AWARDS

Certified SolidWorks Associate (CSWA) May 2024

Dassault Systèmes

Provincial Second Prize, National Undergraduate Mechanical Innovation Design Competition (test sample) Nov 2023

Organizing Committee